

Mathematics for Computing

Tutorial 01

Dasuni Gunasekara dasuni.g@nsbm.ac.lk

Faculty of Computing

NSBM Green University

Law of Indices

Q1) Simplify the following:

1.1

- i. $(n^3)^6$
- ii. $(b^{-7})^3$
- iii. $(3a)^4$
- iv. $(9x^5)^2(x^2)^5$
- v. $(4 \times 10^5)^2$
- vi. $(2 \times 10^{-3})^5$
- vii. $(n^8)^4$
- viii. $(n^4)^8$
- ix. $(c^2)^5$
- x. $(q^{10})^{10}$

1.2

- i. $(w^7)^{-1}$
- ii. $(x^3)^{-5}$
- iii. $d(d^{-2})^{-9}$
- iv. $(z^8)^0 z^5$
- v. $(a^5)^3 c^4$
- vi. $(c^3)^5 (d^3)^0$
- vii. $(t^2)^{-2} (t^2)^{-5}$
- viii. $(m^3)^{-1} (x^2)^5$

Q2).

2.1 Find the area of a square whose side is $5x^3$

2.2 Write an expression for the perimeter of the above square

2.3 Find the volume of a cube whose side is $2x^2$

Q3) Simplify the following. Express each expression in positive exponents.

i. $(n^5)^2(4mn^{-2})$

ii. $(x^{-2})^2(3xy^5)^4$

iii. $(3c^5)^4(c^2)^3$

iv. $(6ab)^3(5a^{-3})^2$

Q4) Show that $\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n$ for nonzero numbers a and b and positive integers n .

Q5) Simplify each expression:

i. $\left(\frac{a}{5b}\right)^{-2}$

ii. $\left(\frac{2x^6}{y^4}\right)^{-3}$

iii. $\left(\frac{m}{n}\right)^{-3}$

iv. $\left(\frac{3x^2}{5y^4}\right)^{-4}$

Q6) Simplify each expression:

i. $\frac{3^8}{3^6}$

ii. $\frac{3^6}{3^8}$

iii. $\frac{d^{14}}{d^{17}}$

iv. $\frac{n^{-1}}{n^{-4}}$

v. $\frac{5a^{-7}}{10a^{-9}}$

vi. $\frac{x^{11}y^3}{x^{11}y}$

vii. $\frac{c^3d^{-5}}{c^4d^{-1}}$

viii. $\frac{10m^6n^3}{5m^2n^7}$

ix. $\frac{m^3n^2}{m^{-1}n^3}$

x. $\frac{a^{-3}}{a^{-9}}$

Q7) Simplify each expression:

i. $\frac{3^2 m^5 t^6}{3^5 m^7 t^{-5}}$

ii. $\frac{x^5 y^{-8} z^3}{x y^{-4} z^3}$

iii. $\frac{12 a^{-1} b^6 c^{-3}}{4 a^5 b^{-1} c^5}$